```
[Acceptor-5001] Listening on port 5001
```

```
[Acceptor-5002] Listening on port 5002
```

```
[Acceptor-5003] Listening on port 5003
```

```
[Acceptor-5003] Listening on port 5003
[Acceptor-5003] Received: PrepareMessage
[Acceptor-5003] STATE: promised=0, accepted=0, value=null
[Acceptor-5003] Promised proposal 1
[Acceptor-5003] STATE: promised=1, accepted=0, value=null
[Acceptor-5003] Received: AcceptMessage
[Acceptor-5003] STATE: promised=1, accepted=0, value=null
[Acceptor-5003] Accepted value 'TX-1772910690136: Alice->Bob $100.0' for proposal 1
[Acceptor-5003] STATE: promised=1, accepted=1, value=TX-1772910690136: Alice->Bob $100.0
[Acceptor-5003] Received: PrepareMessage
[Acceptor-5003] STATE: promised=1, accepted=1, value=TX-1772910690136: Alice->Bob $100.0
[Acceptor-5003] Rejected proposal 1 (already promised 1)
[Acceptor-5003] STATE: promised=1, accepted=1, value=TX-1772910690136: Alice->Bob $100.0
```

```
[Acceptor-5002] Listening on port 5002
[Acceptor-5002] Received: PrepareMessage
[Acceptor-5002] STATE: promised=0, accepted=0, value=null
[Acceptor-5002] Promised proposal 1238696487
[Acceptor-5002] STATE: promised=1238696487, accepted=0, value=null
[Acceptor-5002] Received: AcceptMessage
[Acceptor-5002] STATE: promised=1238696487, accepted=0, value=null
[Acceptor-5002] Accepted value 'TX-1772912705260: Alice->Bob $100.0' for proposal 1238696487
[Acceptor-5002] STATE: promised=1238696487, accepted=1238696487, value=TX-1772912705260: Alice->Bob $100.0
[Acceptor-5002] Received: PrepareMessage
[Acceptor-5002] STATE: promised=1238696487, accepted=1238696487, value=TX-1772912705260: Alice->Bob $100.0
[Acceptor-5002] Promised proposal 1238697348
[Acceptor-5002] STATE: promised=1238697348, accepted=1238696487, value=TX-1772912705260: Alice->Bob $100.0
[Acceptor-5002] Received: AcceptMessage
[Acceptor-5002] STATE: promised=1238697348, accepted=1238696487, value=TX-1772912705260: Alice->Bob $100.0
[Acceptor-5002] Accepted value 'TX-1772912706121: Bob->Charlie $50.0' for proposal 1238697348
[Acceptor-5002] STATE: promised=1238697348, accepted=1238697348, value=TX-1772912706121: Bob->Charlie $50.0
```

The root cause of the second proposal's failure was the proposer's use of a hardcoded proposal ID of 1 for every proposal. Paxos enforces a strict rule that proposal IDs must be unique and monotonically increasing — each new proposal must carry a higher number than any previously used. During the first round, the acceptors received Prepare(ID=1), recognised it as greater than their initial promised value of zero, and responded with a Promise before accepting the value "Alice→Bob $100.0". Once that promise was made, the acceptors committed to rejecting any future proposal carrying the same or a lower ID. So when the second proposal arrived with the identical ID of 1, all three acceptors rejected it outright, and the proposal collapsed before even reaching Phase 2.

The fix was to replace the static FIXED_PROPOSAL_ID = 1 with a dynamically generated value combining the current system timestamp with an incrementing counter. This produced strictly increasing IDs — 1238696487 for the first proposal and 1238697348 for the second —

guaranteeing that each new proposal is always distinguishable and numerically greater than the last. With these unique IDs in place, both acceptors successfully promised and accepted both proposals, committing "Alice→Bob $100.0" and "Bob→Charlie $50.0" without conflict. This exercise demonstrates that monotonically increasing proposal numbers are not optional in Paxos — they are a core safety requirement that prevents stale or duplicate proposals from disrupting consensus.

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000004
👑 New leader elected: /election/node-0000000004
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000005
👀 Running as follower
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000006
👀 Running as follower
```

```
Submitted: /payments/payment-0000000002 → alice->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000004
👑 New leader elected: /election/node-0000000004
[Leader] Committed payment: alice->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000005
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000006
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
```

```
Submitted: /payments/payment-0000000003 → bob->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000004
👑 New leader elected: /election/node-0000000004
[Leader] Committed payment: alice->mark:$100
[Leader] Committed payment: bob->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000005
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
[Follower] Applied payment: bob->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000006
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
[Follower] Applied payment: bob->mark:$100
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000005
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
[Follower] Applied payment: bob->mark:$100
👑 New leader elected: /election/node-0000000005
```

```
Submitted: /payments/payment-0000000004 → diana->mark:$60
```

```
🆔 Node ID: /election/node-0000000005
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
[Follower] Applied payment: bob->mark:$100
👑 New leader elected: /election/node-0000000005
[Leader] Committed payment: diana->mark:$60
```

```
⏳ Starting payment node...
🆔 Node ID: /election/node-0000000006
👀 Running as follower
[Follower] Applied payment: alice->mark:$100
[Follower] Applied payment: bob->mark:$100
[Follower] Applied payment: diana->mark:$60
```